EABS & POS Conceptual Design Report

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1 Introduction

This report presents the preliminary conceptual design of the Extended Automated Bottling System (EABS) and its integration with the Purchase Order System (POS), developed for Advantech Ltd. Advantech manufactures high-value custom liquid products that are mixed to a customer-specified recipe and bottled under strictly controlled conditions. The system we are designing automates the full production pipeline, from acceptance of a customer purchase order through to sorted storage of finished, labelled bottles, using the SystemJ design methodology and a Controller/Plant separation.

Our design extends the core Automated Bottling Station (ABS) with three additions, each developed by one team member as their individual project (IP). These include fault tolerance features that detect failures in the production line at run time and mitigate them through redundancy and safe re-routing of workpieces, digital twins of workpieces and workstations that preserve a product's identity and history through production and into its after-sales life, and a Big-Picture Visualisation that gives a system-wide view of the whole production line. This report describes the overall system architecture and control strategy, the core ABS/POS functionality, how each IP integrates with the group project, the allocation of tasks between group members, and the progress made so far, including the completed model of the two-liquid filling machine for the interim.

2 System Overview and Architecture

The design follows the Controller/Plant structure where every intelligent machine is represented by a Controller Model, which implements the machine's control algorithm in the same form that will eventually run on the embedded platform of the real machine, and a Plant Model, which simulates the physical behaviour of the corresponding sensors and actuators. The pipeline currently consists of seven controller/plant pairs, matching the physical flow of the bottle through the station: Loading Conveyor Belt, Rotary Table, Filling Unit, Lid Placing, Cap Screwing, Labeller, and Unloader/Sorting.

Above the machine layer is a System Coordinator, which receives purchase order parameters from the POS and distributes them to the relevant controllers, groups telemetry and fault information from every controller, and decides the destination of each bottle at the sorting stage. The Coordinator also feeds state information to a Digital Twin module and a Big-Picture Visualisation module, both of which are outputs of individual projects described in Section 7. The overall architecture is shown in Figure 1.

Each Controller Model and its paired Plant Model run as an independent SystemJ clock domain, communicating with its neighbours over point-to-point signals rather than shared memory. This keeps a station's internal logic self-contained, a fault or delay inside one station's synchronous reactions cannot corrupt another station's state directly, it can only become visible through a signal that station chooses to emit. The System Coordinator sits above this layer as its own clock domain, exchanging a smaller set of signals with each station, order parameters going down and status and telemetry coming up, rather than participating in every station's internal reactions. This separation is also why the individual projects can each attach to the architecture independently, the Big-Picture Visualisation and Digital Twin modules both consume the same status and telemetry signals the Coordinator already collects, and fault tolerance adds new signals of the same kind rather than a different communication mechanism.

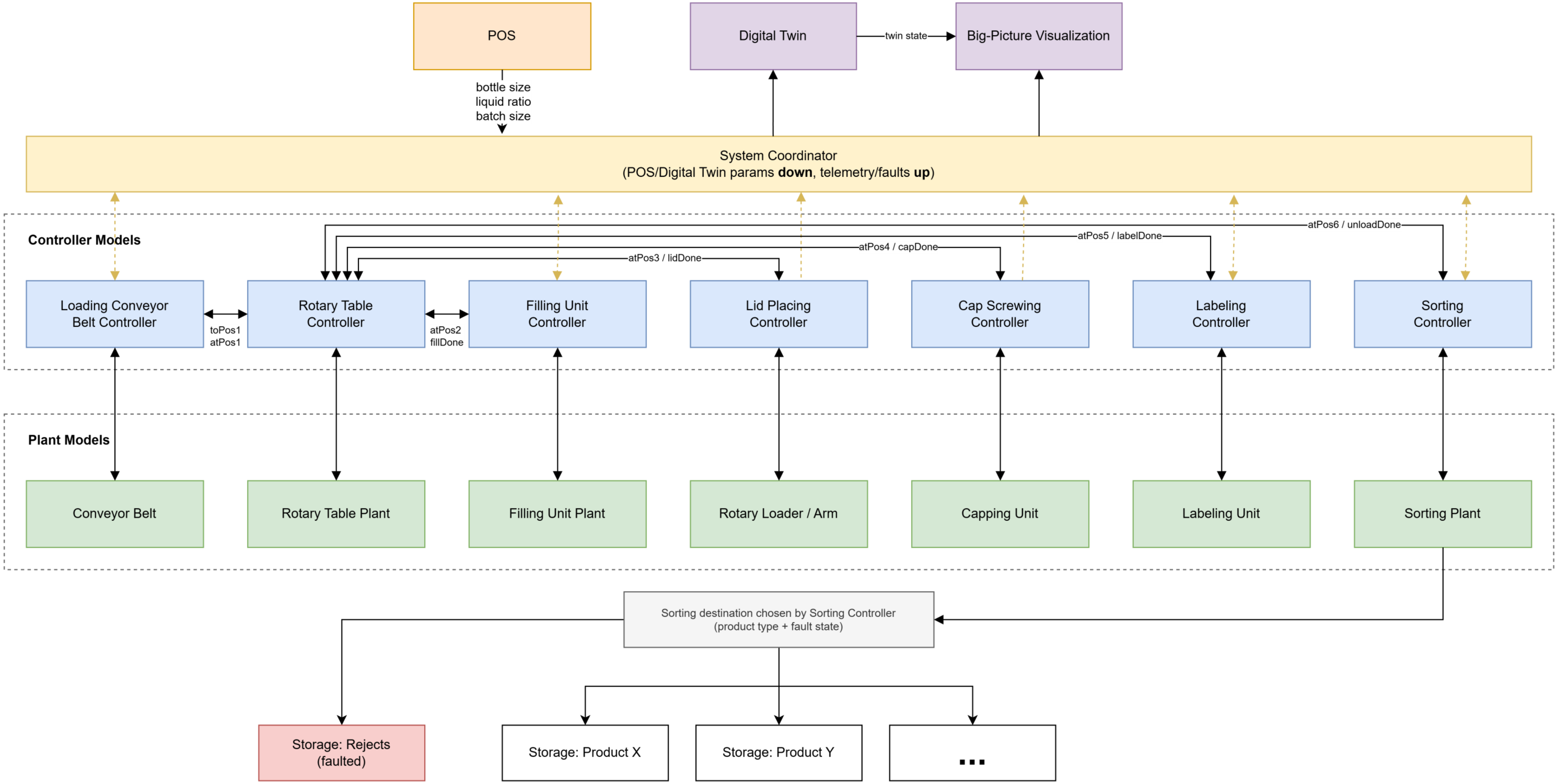

Figure 1. High-level conceptual architecture of the EABS and POS.

3 Control Strategy

We have considered three control strategies for coordinating the machine pipeline: a fully centralised model, in which a single coordinator makes every scheduling and handoff decision, a fully decentralised model, in which each controller communicates directly with its neighbours, and a mixed model. Our team decided to adopt the mixed control strategy that consists of both centralised and decentralised elements.

Deciding where to route a finished bottle, whether a batch is completed, or how to respond to a fault raised anywhere in the line needs a single, consistent view of the whole system. A single System Coordinator is responsible for these system-wide decisions. It pushes purchase order parameters (bottle size, liquid ratio, batch size) down to the relevant controllers and sends status data to the Digital Twin and Big-Picture Visualisation modules. Handing a bottle from one workstation to the next, in contrast, needs to happen every cycle and needs to be fast. Routing this through a central coordinator on every tick would add latency and create a single point of contention for no benefit. Therefore, neighbouring controllers exchange short update signals directly, for example the atPos1 through atPos6 signals and the fillDone, lidDone, capDone, labelDone, and unloadDone signals shown in Figure 1. This split gives the Coordinator authority over the decisions that genuinely need a system-wide view, while keeping each controller's per-cycle logic simple, local, and independently testable.

4 Core ABS Description

Empty bottles are made available at the loading end of the input conveyor. The Loading Conveyor Belt Controller drives the conveyor until a bottle reaches the turntable and hands control to the Rotary Table Controller. The Rotary Table Controller rotates the turntable in 60 degree steps between the five active positions (loading/unloading, filling, lid placement, capping, and the collection point), waiting at each position for the relevant workstation to signal completion before advancing.

At the filling position, the Filling Unit Controller receives the target liquid ratio and total volume for the current batch from the System Coordinator and doses two liquids sequentially through separate inlet and injection valves. This unit is the subject of our completed interim model, detailed in Section 6.

At the lid placing position, the Rotary Loader/Arm supplies a lid from its magazine and places it on the bottle. The Cap Screwing Controller then clamps the bottle, lowers a gripper to pick up the lid, twists it onto the bottle, and retracts. Once capped, the bottle advances to the Labeller, our extension to the core ABS, which prints and applies a label carrying the product identifier, recipe summary, and completion time before the bottle returns to the conveyor.

Finally, the Unloader/Sorting Controller removes the finished bottle from the conveyor. The destination is chosen by the System Coordinator based on the product type recorded for that batch and is communicated back to the Unloader/Sorting Controller for execution. Presence sensors at the loading point, the turntable, and each of the five active positions give the Rotary Table Controller and the Coordinator a consistent view of where each bottle is at any point in the cycle, which is the same information the Big-Picture Visualisation and Digital Twin modules read to build their own view of the line.

5 Purchase Order System (POS)

The POS presents a purchase-order form through which a customer specifies bottle size, the ratio of the liquids to be mixed, and the batch volume. On submission, the POS transmits these parameters to the System Coordinator, which treats them as the active recipe for the next production batch. Batches must complete fully before the next one starts, since reconfiguring the line mid-batch would mix recipes together, so the Coordinator only accepts a new set of parameters from the POS once it has been confirmed the current batch has finished and the line has been reconfigured for the next recipe. The POS subsequently receives a completion notification from the Coordinator so the customer can be informed of the order status.

6 Interim Progress: Filling Unit Model

For the interim, we completed the controller and plant model of the Filling Unit, shown in Figure 2, together with a first version of its visualisation (FillerGUI).

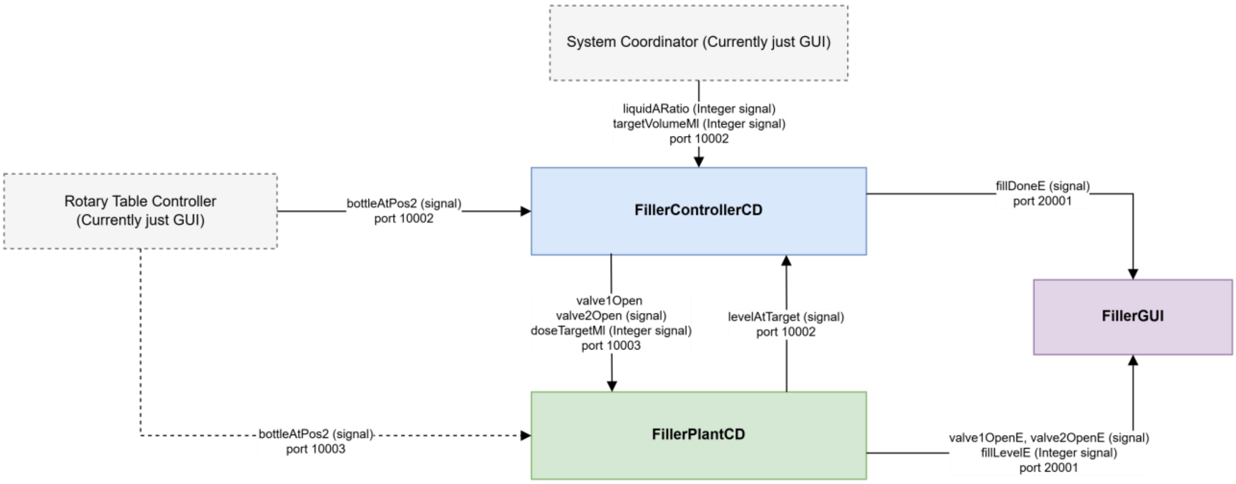

Figure 2. Filling Unit controller/plant model and interface signals.

The Filling Unit uses our proposed structure of a controller and a plant. FillerControllerCD receives liquidARatio and targetVolumeMl (both integer signals) from the System Coordinator, together with the bottleAtPos2 presence signal from the Rotary Table Controller, and outputs valve1Open, valve2Open, and doseTargetMl to FillerPlantCD, which models the two injection valves and the fill-level sensor. FillerPlantCD reports levelAtTarget back to the controller once the correct volume has been dispensed, and the controller signals fillDoneE back to the Coordinator so the Rotary Table can advance the bottle. The FillerGUI subscribes to the plant's valve and fill-level events and renders the current fill state for the operator. The two liquids are sequenced so that the valve for the second liquid only opens once the first has reached its own target volume, which prevents the two liquids mixing in the wrong proportion or overflowing the bottle.

7 Extended ABS (EABS) Features

Each group member is developing one advanced feature of the EABS as an individual project (IP) that extends the base ABS described in Section 4. The full details of each feature are described in each group member's IP report, summarised here.

7.1 Fault Tolerance (Christian)

The base ABS has no way of detecting a fault inside a station or continuing production if one occurs. This IP adds local fault detection to the Filling Unit, Lid Placement, and Capper, the three stations that actively work on a bottle rather than just transport it. Detection differs by failure mode. The Filling Unit uses a reaction-count timeout to catch a valve left open, since its fault is a continuous condition, while Lid Placement and Capper retry a failed pick-and-place or screwing action a fixed number of times before timing out, since both handle a physical object that can be knocked loose or flung off without the station itself being broken. Once raised, a fault takes the station out of rotation, flags it on its GUI, and reports it to the System Coordinator, which factors it into routing under the fault-response authority it already holds in Section 3, rather than stopping the line outright. An optional extension routes a faulted bottle through the turntable's vacant slot into a fully redundant parallel line, pursued only if the baseline detect-and-flag design proves insufficient once built.

7.2 Digital Twins and After-Sales Tracking (Marie)

Each workpiece is assigned a digital twin when it is loaded onto the line, preserving its identity through every station it passes and into its after-sales life. The twin tracks which recipe was used, which stations processed it and when, and its final storage location, giving a full production history for each bottle rather than only a batch-level record, which also gives the Filling Unit's fault detection and any future station's fault detection a place to record which specific bottle was affected by a fault rather than only which station raised it. Full details, including how the twin's identifier is generated and what happens to it after the bottle leaves storage, are in the individual project report.

7.3 System-Level Big-Picture Visualisation (Tania)

The base ABS lacks high-level observability, making it difficult for operators to monitor plant conditions or diagnose component faults in real time. The Big-Picture Visualisation feature aims to solve this issue by introducing a system-wide interface that displays the whole machine pipeline and the status at each stage. This feature will also include adding the status information to each machine's GUI, which can be accessed from the Big-Picture view.

8 Task Allocation

Member	Group Project Responsibility	Individual Project Topic
Christian Motta	Filling Unit, POS, System Coordinator POS parameters	Fault Tolerance
Marie Patlong	Capper, Sorting Unit, Labeling Unit	Digital Twins
Tania Pan	Rotary Table, Conveyer Belt, System Coordinator Telemetry	System-Level Big-Picture Visualisation

Tasks were allocated during our initial group meeting, based on each member's proposed individual project topic, with each member taking responsibility for building the controller and plant model of their assigned station in the group project in addition to developing their individual project. Each member is also responsible for that station's GUI, since the visualisation for a station is easiest to build and test alongside the plant model it is displaying. Interim progress has focused on the Filling Unit, since it is the station furthest along and gives the clearest example for the System Coordinator's signal interface once that is designed, with the remaining stations and each member's individual project continuing in parallel toward Milestone 2.

9 Assumptions

At most two liquids are mixed per bottle in this iteration of the model.

A bottle's size and target volume are fixed for the duration of a batch and are supplied once per batch by the POS via the System Coordinator, rather than being re-identified per bottle by a camera or barcode reader.

Human-presence and environmental sensors are assumed available at the ABS boundary. If a person is detected in the zone, or if temperature, humidity, or light readings fall outside configured limits, the System Coordinator halts production and resumes once all in-progress bottles are cleared from the line before the batch continues.

Workpiece digital twin identity is assigned when a bottle is loaded onto the line, using an incrementing identifier rather than a serial number or barcode scheme.

The scope of fault detection and mitigation for each station, including which failure modes are covered, is described in the relevant individual project report rather than repeated here.

At most one bottle occupies each of the turntable's active positions at a time, and the conveyor is assumed to have enough capacity that a bottle is never blocked from advancing by a full downstream buffer, so the model does not need to reason about queuing or backpressure along the line.

Deadlock is not treated as a design risk that needs its own avoidance mechanism, since SystemJ's synchronous reaction model rules it out by construction for the kind of signal exchange used here.

10 Milestone 2 Plan

Between now and Milestone 2, we will complete the remaining machines, which includes each of their controller/plant pairs as well as their GUIs. We will also implement the System Coordinator logic, integrate each member's IP (Big-Picture Visualisation and GUI extension, fault detection, and digital twins) into the final model, and complete the POS to display order status from the Coordinator.

Once every station is modelled, we will run the full pipeline end to end with a representative purchase order to confirm bottles move through all seven stations correctly, that the Coordinator's batch and sorting decisions match the recipe supplied by the POS, and that each IP's contribution is visible in the running system rather than only in isolation, a faulted station rerouting correctly, a digital twin's history being retrievable for a finished bottle, and the Big-Picture view reflecting the state of every station at once. This end-to-end run is also what the final demonstration and interview will be based on.